Effective heights of sums of elliptic abscissas

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Abstract

For an elliptic curve $y^2 = x^3 - Dx$ over $\mathbb{Q}(\sqrt{2})$ with integral D, we bound the height lost when adding the abscissas of finitely many integer multiples of a fixed point. If the abscissas are positive at both real embeddings, the loss is at most an explicit constant times the logarithm of the largest index. The proof uses formal logarithms only as identities: it requires no lower bound for a transcendental linear form. Relative to any individual abscissa, for four multiples the loss is at most $24 \log B + \log 864$. Applied to the fixed elliptic source, this gives a numerical bound in terms of the two rational components of the sum, valid for all nonzero integer indices.

1 Heights and the main statement

Put $K = \mathbb{Q}(\sqrt{2})$. At a finite place v above a rational prime p, let ord_v have value group $\mathbb{Z}$, let f_v be the residue degree, and put $\lambda_v = f_v \log p/2$. Thus $\mathrm{ord}_v(p) = e_v$ and $\sum_{v|p} e_v f_v = 2$. For $a_1, \dots, a_r \in K$ define

$$h(1, a_1, \dots, a_r) = \sum_{v \nmid \infty} \lambda_v \max(0, -\mathrm{ord}_v(a_1), \dots, -\mathrm{ord}_v(a_r)) + \frac{1}{2} \sum_{\sigma : K \hookrightarrow \mathbb{R}} \log \max(1, |\sigma(a_1)|, \dots, |\sigma(a_r)|).$$

For one entry this is the absolute logarithmic Weil height $h(a) = h(1, a)$.

Let $D \in \mathbb{Z}[\sqrt{2}] \setminus \{0\}$ and

$$E_D : \quad y^2 = x^3 - Dx.$$

Let $P \in E_D(K)$ have infinite order. For nonzero integers $n_1, \dots, n_r$ set

$$x_i = x([n_i]P), \qquad Z = \sum_{i=1}^{r} x_i, \qquad B = \max_i |n_i| \geq 1.$$

The points $[n_i]P$ are neither the origin nor a point of order two, so every x_i is defined and nonzero.

Theorem 1. *Assume $\sigma(x_i) > 0$ at both real embeddings of K. Put*

$$A_r = (r-1)2^{r-1}, \qquad B_r = \sum_{k=1}^{r} \binom{r-1}{k-1} \log k.$$

Then

$$h(Z) \geq \max_i h(x_i) - A_r \log B - B_r. \tag{1}$$

In particular, for $r = 4$,

$$h(Z) \geq \max_i h(x_i) - 24 \log B - \log 864. \tag{2}$$

There is also a comparison with the joint coordinate height. Put

$$C_r = 2r + 2\sum_{k=1}^{r}\binom{r}{k}(k-1) = (r-2)2^r + 2r + 2, \qquad D_r = \sum_{k=1}^{r}\binom{r}{k}\log k.$$

$$h(1, x_1, \dots, x_r) - C_r \log B - D_r \;\le\; h(Z) \;\le\; h(1, x_1, \dots, x_r) + \log r. \tag{3}$$

All constants in these bounds are independent of D, P, and the indices, subject to the stated hypotheses.

The positivity hypothesis is automatic when $P \in 2E_D(K)$. Indeed the duplication formula is

$$x(2Q) = \left(\frac{x(Q)^2 + D}{2y(Q)}\right)^2. \tag{4}$$

For a point Q of infinite order the displayed denominator and numerator are nonzero: otherwise $2Q$ would be the origin or $(0,0)$. Hence its image under each real embedding is a strictly positive square.

2 An integral local principal part

We give the local calculation at every finite place, including places of bad reduction. Put $t = -x/y$, the parameter at the origin. The integral Weierstrass equation defines its usual formal group over each valuation ring. Its points with t in the maximal ideal form an open subgroup $E_1(K_v)$; no good-reduction hypothesis is needed at the smooth origin. For a finite nonzero point, membership in $E_1(K_v)$ is equivalent to $\operatorname{ord}_v(x) < 0$. In that case

$$\operatorname{ord}_v(x) = -2\operatorname{ord}_v(t), \qquad \operatorname{ord}_v(y) = -3\operatorname{ord}_v(t). \tag{5}$$

For example, these equalities follow directly by comparing the terms of $y^2 = x^3 - Dx$ when $\operatorname{ord}_v(x) < 0$.

Lemma 2. *Let v be any finite place of K. The formal logarithm ℓ_v on $E_1(K_v)$ satisfies, for $a = \operatorname{ord}_v(t) > 0$,*

$$\operatorname{ord}_v(\ell_v(t)) = a, \qquad x(t) - \ell_v(t)^{-2} \in \mathcal{O}_v.$$

It is injective on $E_1(K_v)$.

Proof. The equation in the parameter t gives

$$x(t) = \frac{1 + \sqrt{1 + 4Dt^4}}{2t^2} = t^{-2} + Dt^2 - D^2t^6 + 2D^3t^{10} - \cdots. \tag{6}$$

The coefficients after t^{-2} are signed Catalan numbers times powers of D, and are integral. Thus $x(t) - t^{-2} \in \mathcal{O}_v$.

The invariant differential and its integral are

$$\frac{dx}{2y} = (1 + 4Dt^4)^{-1/2}\,dt, \tag{7}$$

$$\ell_v(t) = \sum_{k\ge 0}\frac{(-1)^k\binom{2k}{k}D^k}{4k+1}\,t^{4k+1}. \tag{8}$$

If $v \mid 2$, every denominator $4k+1$ is a unit; the terms with $k \ge 1$ have valuation at least $5a$. If $v \nmid 2$, the extension $K/\mathbb{Q}$ is unramified at v. For $k \ge 1$ and odd p one has

$$v_p(4k+1) \le 4k - 2,$$

because $3^{4k-2} \geq 4k+1$ (by induction, starting with $9 \geq 5$). The k-th term of (8) consequently has valuation at least

$$(4k+1)a - v_p(4k+1) \geq 3a.$$

Moreover $v_p(4k+1) \leq \log_p(4k+1) = o(k)$, so the valuations of the terms tend to infinity. At places above two this also follows immediately from integrality of the coefficients. Hence at every finite place $\operatorname{ord}_v(\ell_v(t) - t) \geq 3a$, and in particular $\operatorname{ord}_v(\ell_v(t)) = a$. Writing $\ell_v(t) = t(1+\eta)$ with $\operatorname{ord}_v(\eta) \geq 2a$ gives

$$\ell_v(t)^{-2} - t^{-2} = t^{-2}\frac{-2\eta - \eta^2}{(1+\eta)^2} \in \mathcal{O}_v.$$

Together with (6), this proves the claimed principal part. The integral of the invariant differential is a formal-group homomorphism. The valuation equality shows that its only zero in $E_1(K_v)$ is the origin, which proves injectivity. □

Since $E_D(K_v)$ is compact and $E_1(K_v)$ is an open subgroup, their quotient is finite. Choose $m > 0$ with $[m]P \in E_1(K_v)$ and define

$$L_v = \frac{\ell_v([m]P)}{m}.$$

The formal-group homomorphism shows this is independent of the choice of m. It is nonzero, since P has infinite order and Lemma 2 is injective. Whenever $[n]P \in E_1(K_v)$,

$$\ell_v([n]P) = nL_v, \qquad x([n]P) = \frac{1}{n^2 L_v^2} + \varepsilon_n, \quad \varepsilon_n \in \mathcal{O}_v. \tag{9}$$

This identity is the only use of a local logarithm in the height argument.

3 Summing the local losses

For each nonempty subset $I \subseteq \{1, \dots, r\}$ put

$$T_I = \sum_{i \in I} \frac{1}{n_i^2} \in \mathbb{Q}_{>0}.$$

Write $T_I = A_I/B_I$ in lowest terms, with positive integers A_I, B_I.

Lemma 3. *Fix an index i. At every finite place where $[n_i]P$ is formal, let J be the set of all formal indices and put*

$$C_J = \sum_{j \in J} \prod_{k \in J \setminus \{j\}} n_k^2 \in \mathbb{Z}_{>0}.$$

The loss of the local height of x_i upon passing to Z is at most $\operatorname{ord}_v(C_J)$.

Proof. Write $M_i = -\operatorname{ord}_v(x_i) > 0$. The principal-part identity is

$$Z = (n_i L_v)^{-2} A_{J,i} + \varepsilon, \qquad A_{J,i} = n_i^2 T_J = \frac{C_J}{\prod_{j \in J \setminus \{i\}} n_j^2}, \qquad \varepsilon \in \mathcal{O}_v.$$

If $\operatorname{ord}_v(A_{J,i}) < M_i$, the local height of Z is $M_i - \operatorname{ord}_v(A_{J,i})$; otherwise it is zero. The loss is therefore at most $\max(0, \operatorname{ord}_v(A_{J,i}))$. All the indices are integers, so $\operatorname{ord}_v(A_{J,i}) \leq \operatorname{ord}_v(C_J)$; the latter is nonnegative. □

Lemma 4. *At a finite place v, let*

$$M_v = \max(0, -\operatorname{ord}_v(x_1), \dots, -\operatorname{ord}_v(x_r)), \quad m_v = \max(0, -\operatorname{ord}_v(Z)).$$

If $M_v > 0$, let I_v be the nonempty set of indices whose points belong to $E_1(K_v)$, and choose $j \in I_v$ with $\operatorname{ord}_v(n_j) = \max_{i \in I_v} \operatorname{ord}_v(n_i)$. Then

$$0 \le M_v - m_v \le \operatorname{ord}_v(n_j^2 T_{I_v}) \le 2 \max_i \operatorname{ord}_v(n_i) + \max(0, \operatorname{ord}_v(T_{I_v})).$$

Proof. The remaining abscissas are integral at v. By (9),

$$Z = \frac{T_{I_v}}{L_v^2} + \varepsilon, \qquad \varepsilon \in \mathcal{O}_v, \qquad M_v = 2 \operatorname{ord}_v(n_j L_v) > 0.$$

The number $c = n_j^2 T_{I_v} = \sum_{i \in I_v} (n_j/n_i)^2$ is nonzero and integral at v. If $\operatorname{ord}_v(c) < M_v$, the principal part has negative valuation and $m_v = M_v - \operatorname{ord}_v(c)$. Otherwise the principal part and Z are integral, and $m_v = 0$. Thus the loss is $\min(M_v, \operatorname{ord}_v(c))$, proving the inequalities. □

Proof of Theorem 1. Fix an index i. Where $[n_i]P$ is not formal, its finite local height is zero and there is no positive loss. At every other place Lemma 3 applies. As J ranges over the subsets containing i, the sum of its contributions is at most

$$\sum_{J \ni i} \sum_{v \nmid \infty} \lambda_v \operatorname{ord}_v(C_J) = \sum_{J \ni i} \log C_J.$$

For $|J| = k$ one has $C_J \le k B^{2(k-1)}$, and there are $\binom{r-1}{k-1}$ such subsets. Consequently this total is at most

$$2 \sum_{k=1}^{r} \binom{r-1}{k-1} (k-1) \log B + B_r = (r-1) 2^{r-1} \log B + B_r.$$

At a real embedding the positive sum is at least its i-th summand, so there is no archimedean loss. This proves (1) for every i. The values $A_4 = 24$ and $B_4 = 3\log 2 + 3\log 3 + \log 4 = \log 864$ give (2).

The ultrametric inequality shows $m_v \le M_v$. At places with $M_v = 0$ there is no loss. Sum Lemma 4, using its last inequality, to obtain

$$\sum_{v \nmid \infty} \lambda_v (M_v - m_v) \le 2 \sum_{i=1}^{r} \log |n_i| + \sum_{\varnothing \ne I \subseteq \{1,\dots,r\}} \log A_I. \tag{$*$}$$

Here, for any positive rational $a = A/B$ in lowest terms,

$$\sum_{v \nmid \infty} \lambda_v \max(0, \operatorname{ord}_v(a)) = \sum_p \log p \, \max(0, v_p(a)) = \log A.$$

Also $\max_i \operatorname{ord}_v(n_i) \le \sum_i \operatorname{ord}_v(n_i)$ because all the n_i are integers. This proves $(*)$, even though the subset I_v varies with the place: each of the finitely many subsets contributes at most its full positive-valuation sum.

If $|I| = k$, a common denominator for T_I gives

$$A_I \le \sum_{i \in I} \prod_{j \in I \setminus \{i\}} n_j^2 \le k B^{2(k-1)}.$$

Together with $\log |n_i| \le \log B$, this bounds $(*)$ by $C_r \log B + D_r$. At each real embedding all x_i are positive, so

$$\max(1, \sigma(x_1), \dots, \sigma(x_r)) \le \max(1, \sigma(Z)) \le r \max(1, \sigma(x_1), \dots, \sigma(x_r)).$$

Combining the finite and infinite places proves (3). Its four-coordinate constants are $C_4 = 42$ and $D_4 = \log 20736$. □

4 The fixed arithmetic source

Take

$$D = 11 - 6\sqrt{2} = (3 - \sqrt{2})^2, \qquad R_0 = (1 + \sqrt{2}, 2\sqrt{2}),$$
$$P = 2R_0 = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

The equation and the doubling identity are checked directly. At the place $\pi = \sqrt{2}$ one has $\mathrm{ord}_\pi(x(P)) = -6$ and $\mathrm{ord}_\pi(y(P)) = -9$; hence $\mathrm{ord}_\pi(t(P)) = 3$. Lemma 2 proves that $\ell_\pi(P) \neq 0$, so P has infinite order. Formula (4) proves the positivity needed in Theorem 1 for every nonzero integer index.

The arithmetic source uses the index-height assertion

$$h(x([n]P)) \geq n^2 - 6 \qquad (n \in \mathbb{Z} \setminus \{0\}). \tag{10}$$

We state the following application explicitly under that retained assertion; its precise source identifier is `eq:o8-indexheight`, as documented in [2]. The sum-height theorem above is independent of this assertion.

Corollary 5. *Assume* (10). *For nonzero integers* n_1, n_2, n_3, n_4, *put*

$$Z = \sum_{i=1}^{4} x([n_i]P) = S_0 + S_1\sqrt{2}, \qquad S_0, S_1 \in \mathbb{Q}, \qquad B = \max_i |n_i|.$$

Then

$$B^2 \leq 2h(Z) + 80 \leq 2h(S_0) + 2h(S_1) + 83. \tag{11}$$

No restriction on the parity of the indices is imposed.

Proof. Theorem 1 gives

$$h(Z) \geq B^2 - 24\log B - 6 - \log 864.$$

For $B > 0$, the inequality $\log t \leq t - 1$ with $t = B^2/24$ gives

$$24\log B - \frac{B^2}{2} \leq 12\log 24 - 12 < 27.$$

Since $\log 864 < 7$, the total subtracted constant is less than 40. Thus $h(Z) \geq B^2/2 - 40$. Finally the elementary height inequalities for addition and multiplication give

$$h(S_0 + S_1\sqrt{2}) \leq h(S_0) + h(S_1) + \tfrac{3}{2}\log 2.$$

Because $3\log 2 < 3$, the second inequality follows after increasing the additive constant from 80 to 83. The statements $\log 24 < 13/4$ and $\log 864 < 7$ can, if desired, be verified by finite positive Taylor sums for the exponential; exact certificates are supplied with this paper. □

The two rational components S_0, S_1 are linear combinations of the existing abscissa components. Thus this aggregate requires no multiplication coordinates. Inequality (11) is the height input for a construction retaining the three scales 1, S_0^2, and $(1 + S_1^2)^2$. It does not assert an effective bound in terms of $h(S_0)$ alone; taking the field trace can introduce additional cancellation between conjugate blocks.

5 A number-field extension

Corollary 6. *Let F be a totally real number field unramified at every odd rational prime. Let $D \in \mathcal{O}_F \setminus \{0\}$ and let $P \in 2E_D(F)$ have infinite order. Then both inequalities (1) and (3) hold for the absolute heights over F, with the same constants A_r, B_r, C_r, D_r.*

Proof. At every odd place the valuation estimate in Lemma 2 uses only that the ramification index is one. At places above two, all coefficients of the formal logarithm are integral, regardless of the ramification index. The lemma therefore holds over F at every finite place. In the product-formula calculation replace $f_v \log p/2$ by $f_v \log p/[F : \mathbb{Q}]$; the contribution of a rational positive integer remains its usual logarithm. At real embeddings positivity follows from doubling as before. The two proofs are otherwise unchanged. □

6 Verification data

The companion program `check_effective_elliptic_sums.py` uses exact rational arithmetic. It verifies the curve and doubling identities, the stated valuation of P, the first twelve coefficients of the formal identities, the combinatorial constants, and finite rational certificates for the two exponential inequalities used above. The local estimates and the summation over all places are proved in the text, rather than inferred from a finite computation.

References

[1] J. H. Silverman, *The Arithmetic of Elliptic Curves*, second edition, Graduate Texts in Mathematics 106, Springer, 2009. Chapters IV and VII supply the standard formal-group and local-group terminology used here. https://doi.org/10.1007/978-0-387-09494-6.

[2] *Arithmetic dependencies of the optimized rational constructions*, 8 November 2026, Section 2, `arithmetic_dependencies_20261108_v8.tex`, project replication archive.